

Moein Razavi, Ph.D.

Senior AI/ML Engineer · Medical Device Software & Clinical AI · Deep Learning & Image Processing

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linkedin.com/in/moein-razavi · github.com/moeinrazavi · moeinrazavi.github.io/moeinrazavi · Scholar (470+ citations)

Professional Summary

Senior AI/ML engineer building and validating deep learning systems where correctness and safety matter. Shipped medical device software at CereVu Medical — wearable diagnostic patches, on-device clinical inference, **400+ patients across 5 clinical studies** — and now build production ML platforms at Ford. **Ph.D.** plus **M.S. Computer Science** and **B.S. Mechanical Engineering** (4.0 GPA, Texas A&M). Fluent in **Python and PyTorch**, with **6+ years of image processing and deep neural networks** plus **rigorous statistical validation** (grouped cross-validation, ROC-AUC, uncertainty quantification). **16 peer-reviewed publications** (470+ citations) — practiced at turning complex systems into precise written documentation. Daily user of **agentic AI tools** (Claude Code, Codex, Cursor). Eager to go deep on FDA requirements, risk management, V&V, and design history files. **U.S.-based** (McKinney, TX), remote since 2023, able to travel.

Technical Skills

Deep Learning & Image Processing: PyTorch (Expert) · TensorFlow · Hugging Face · CNNs · U-Net & Semantic Segmentation · Object Detection (YOLO, Faster R-CNN) · Vision & Sequence Transformers (self-/cross-attention) · Transfer Learning · Focal Loss & Class Imbalance · OpenCV · Core ML On-Device Inference

Medical Device & Clinical AI: Wearable diagnostic device software (hardware-to-cloud) · Embedded clinical decision algorithms · Clinical study data curation & lineage · Biosignal processing (EEG/ERP, PPG/HRV, sensor fusion) · Implantable MEMS microelectrode design & FEA verification · Explainable AI (SHAP, LIME)

Statistics, Validation & Documentation: Grouped k -fold cross-validation · ROC-AUC, sensitivity/specificity, F1 · Hypothesis testing & confidence intervals · Monte Carlo & uncertainty quantification · Failure-mode analysis · Unit/integration/system tests (pytest) · Requirements-to-test traceability · Configuration & release management · Specification writing

Agentic AI, LLMs & RAG: Claude Code · Codex · Cursor · MCP · LangGraph · LangChain · RAG (FAISS, ChromaDB) · Guardrails · Fine-tuning (LoRA/QLoRA)

Languages, Cloud & Infrastructure: Python (Expert) · C++ · Swift/SwiftUI · TypeScript · SQL · MATLAB · FastAPI · REST/WebSocket · React/Next.js · Docker · Terraform · CI/CD (GitHub Actions) · GCP (Vertex AI) · AWS · PySpark

Professional Experience

Senior Data Scientist & AI/ML Engineer

Aug 2023 – Present

Ford Motor Company Detroit, MI (Remote)

- **Own deep learning model development end to end** for safety-critical forecasting: transformers (**TFT, Chronos-2**) in **PyTorch/Hugging Face** with Optuna search on distributed GPUs — **accuracy up 10%, 40% faster training**
- **Quantify and report model uncertainty** for decisions that cannot fail silently: **Monte Carlo simulation (~3,000 scenarios)** with **P50/P90 quantile reporting**, plus backtesting and drift monitoring on every deployment
- **Use agentic AI daily as a productivity multiplier** and ship agentic systems in production: multi-agent platform (**LangGraph**, Vertex AI Gemini) with routing, retrieval (FAISS/ChromaDB), extraction, and **validation agents** — replaced **~500 lines** of brittle parsers with one self-healing, schema-validating pipeline serving **10 U.S. states**
- **Write production-grade, tested Python:** FastAPI services with unit/integration tests, code review, **SonarQube**, Terraform IaC, Tekton CI/CD — **99.9% uptime; 10+ TB** data pipelines (PySpark, BigQuery, Airflow)
- **Communicate results to non-technical audiences:** automated reporting for **500+ stakeholders**, turnaround down **45%**

Jun 2023 – Sep 2024

Software Engineer & AI/ML Researcher (Medical Device Software)

CereVu Medical, Inc. San Francisco Bay Area, CA (Remote)

- **Built end-to-end software for a wearable diagnostic medical device:** firmware-to-cloud pipeline streaming **9 physiological biomarkers** from sensor patches (Nordic UART BLE) through Python microservices to clinician applications, with the **ROPA pain detection algorithm** on the device path — **monitoring precision up 30%**
- **Trained and validated deep learning models** (LSTM, CNN) on clinical biosignal data and **deployed them on-device** (Core ML) for real-time edge inference across **400+ patients in 5 clinical studies** — **15% accuracy improvement**
- **Curated, documented, and versioned the clinical datasets behind every model**, tracking train/validation/test splits so results stayed reproducible and auditable across studies
- **Translated clinical requirements into software specifications and test criteria** with clinicians and study investigators; authored algorithm and interface documentation used by engineering and clinical stakeholders
- **Built Python microservices** (FastAPI, WebSocket) sustaining **10K+ concurrent biosignal streams (40% lower latency)**; shipped **2 iOS apps** and a React/Next.js clinical dashboard under CI/CD with automated test gates

Texas A&M University College Station, TX

- **Designed a dual-branch transformer** (1D-CNN tokenizer, rotary positional encoding, cross-attention fusion, stochastic depth, focal loss) for clinical stress detection from wearable physiological signals (**256 participants, 1.7B rows**); validated by **5-fold grouped cross-validation** at **AUC 0.993 / F1 0.965**, deployed on-device at **60% faster inference**
- **Delivered a production computer vision system** using deep object detection for real-time distancing and mask-compliance monitoring — **98% accuracy** over **50K+ daily interactions (76 citations)**; built a **Conv-LSTM** gait model for fall-risk prediction (**93% accuracy**)
- **Built an 18-feature HRV/PPG signal-processing pipeline** harmonizing **5 heterogeneous clinical datasets** with documented, reproducible preprocessing; applied **explainable AI** (SHAP, LIME) to clinical models and shipped *GlucoseCoach*, an iOS + Flask app for personalized glucose monitoring
- **Created OpenSync** — documented open-source platform synchronizing EEG/ERP, eye-tracking, and biosensor streams, **adopted by 3 labs; synthesized 200+ studies** into a scoping review of clinical ML/DL methods (**79 citations**)

2016 – 2018

Research Assistant & Engineering Team Lead (Implantable MEMS Neural Microelectrodes)

Cognitive Sciences & Technologies Council Tehran, Iran

- **Led the engineering team** designing single, planar, and 3D **implantable neural microelectrodes**: structural **FEA in ANSYS independently cross-validated in COMSOL Multiphysics** — a two-solver verification strategy for a device intended for human neural tissue; published at **MEMS 2018**
- **Built an EEG-based brain-computer interface for robot motion control**: acquisition, filtering, feature extraction (MATLAB DSP, EEGLab), classification, and embedded actuation (ARM/Arduino)

Independent Ventures & Client Work

Founder & Lead Developer

Jan 2023 – Present

InspecaLytics | Clink Social McKinney, TX

- **Scoped requirements directly with clients and shipped two production platforms solo**: a field-inspection suite with offline-first sync and **automated DOCX/PPTX report generation** (documentation cycles down **35%**), plus a real-time social app with secure QR cross-device OAuth

Education

Ph.D., Industrial Engineering (Data Science) | GPA: 4.0/4.0 | Research Fellowship

Aug 2021 – Dec 2023

Texas A&M University, College Station, TX

Dissertation: Advancing Stress Detection — Real-Time, Naturalistic, and Personalized AI Approaches.

M.S., Computer Science | GPA: 4.0/4.0 | Research Fellowship

Aug 2019 – May 2021

Texas A&M University, College Station, TX

Coursework: Deep Learning, Computer Vision, Pattern Recognition, NLP, Software Engineering.

B.S., Mechanical Engineering | Top 1% of ~250,000 Nationwide

2013 – 2018

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

Coursework: Automatic Control, Robotics, FEM, Vibrations, Heat Transfer, CAD/CAM.

Selected Publications (16 peer-reviewed papers · 470+ citations · h-index 10)

Razavi, M., et al. "ML & DL for Stress & Mental Disorder Detection: Scoping Review." *JMIR Mental Health*, 2024. **[79 cites]**

Razavi, M., et al. "Distance & Mask Monitoring via Deep Object Detection." *SN Computer Science*, 2022. **[76 cites]**

Razavi, M., et al. "OpenSync: Synchronizing Measures in Neuroscience." *J. Neuroscience Methods*, 2022. **[31 cites]**

Razavi, M., Nadafi, R. "Design of Single, Planar & 3D Neural Microelectrodes (ANSYS/COMSOL)." *MEMS 2018*.

Certifications & Honors

Generative AI with LLMs – AWS · **Deep Learning Specialization** – DeepLearning.AI · **Machine Learning** – Stanford · **NLP Specialization** – DeepLearning.AI · **Advanced MLOps** – Coursera · **Bayesian Statistics** – Coursera · **Graduate Research Fellowship & Dean's List**, Texas A&M (2019–2023) · **IEEE & HFES** Member